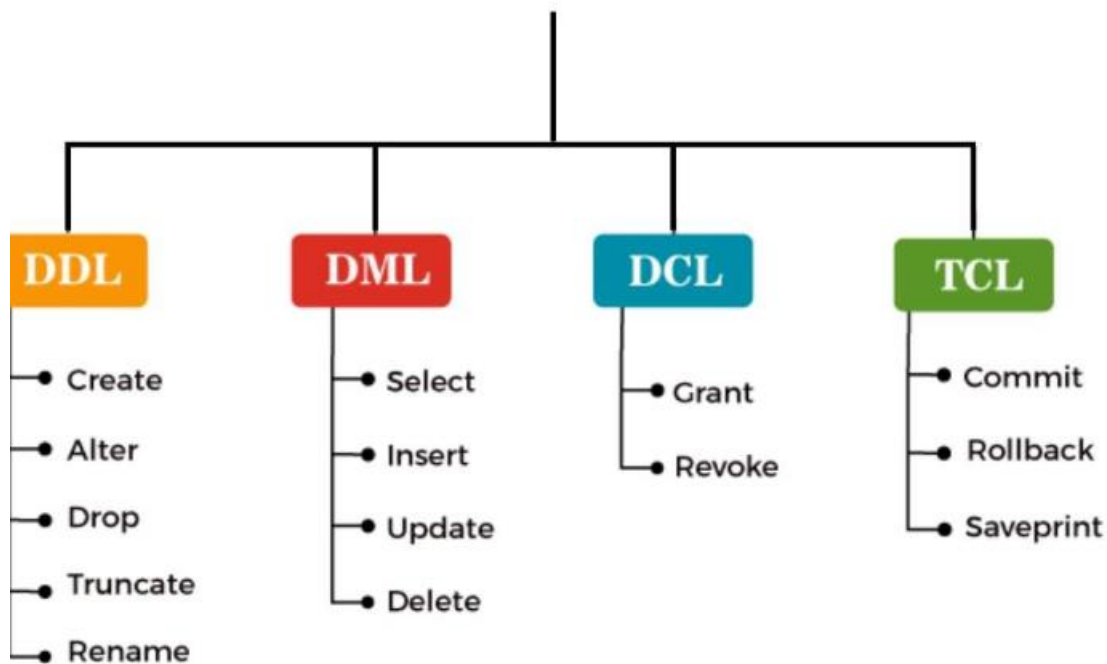


SQL Commands



In MySQL, DDL – Data Definition Language commands are used to define, modify, and remove database structures like databases, tables, indexes, and views.

DDL Command	Purpose
CREATE	Create new DB / table / view
ALTER	Modify table structure
DROP	Remove table/database/view
TRUNCATE	Remove all rows
RENAME	Rename object

create database Anu1;

use Anu1;

show tables;

create table Anu1 (PRN int, Name varchar(10));

insert into Anu1 values(101,"aman");

select * from Anu1;

alter table Anu1 rename to student;

select * from student;

alter table Anu1 add fee decimal(5,2);

alter table student add fee decimal(5,2);

desc student;

alter table student modify fee decimal(6,3);

desc student;

alter table student rename column fee to s_fee;

desc student;

select * from student;

alter table student add fine decimal(5,2);

desc student;

alter table student drop column fine;

desc student;

alter table student add primary key(PRN);

desc student;

alter table student drop primary key;

desc student;

alter table student alter s_fee set default 100;

desc student;

alter table student modify name varchar(10) NOT NULL;

desc student;

alter table student modify name varchar(10);

desc student;

```
SHOW CREATE TABLE student;

ALTER TABLE student MODIFY PRN INT NULL;

desc student;

alter table student modify name varchar(10) int;

alter table student add primary key(PRN);
```

```
SELECT SQRT(25) FROM DUAL;

alter table DUAL rename to st; // we cannot change system table name
```

Drop: Drop Table, Drop database

```
create database S22;

show databases;

use S22;

create table S32(roll int);

show tables;

drop table S32;

show tables;

drop database S22;

show databases;
```

Drop vs Truncate

TRUNCATE

Deletes **all records** from a table but keeps the structure.

```

50 - select * from student;
51 • insert into student values(102,"Raman", 200);
52 ✖ insert into student (PRN, Name) values(102,"Naman"); // see error
53 insert into student(PRN, Name) values(103,"Naman");
54 • select * from student;
55

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [IA](#)

	PRN	name	s_fee
▶	101	aman	NULL
	102	Raman	200.000
	103	Naman	100.000
*	NULL	NULL	NULL

alter table student modify Name varchar(4);

Error Code: 1265. Data truncated for column 'Name' at row 2

Truncate

```

56 • truncate table student;
57 • select * from student;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [IA](#)

	PRN	name	s_fee
*	NULL	NULL	NULL

```

58 • truncate table student;
59 • desc student;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Field	Type	Null	Key	Default	Extra
▶	PRN	int	NO	PRI	NULL	
	name	varchar(10)	YES		NULL	
	s_fee	decimal(6,3)	YES		100.000	

```

65 • create table student2(PRN int);
66 • insert into student2 values(100);
67 • select * from student2;
68 • use Anu1;
69 • show tables;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Tables_in_anu1			
	s2			
	student			
	student2			

```

60 • insert into student values(102,"Raman", 200);
61 • insert into student values(103,"Naman", 300);
62 • select * from student;
63 • insert into student(PRN, Name) values(104,"Rohit");
64 • select * from student;

```

Result Grid		Filter Rows:	Edit:	Export/Import:	Wrap
PRN	name	s_fee			
102	Raman	200.000			
103	Naman	300.000			
104	Rohit	100.000			
NULL	NULL	NULL			

```

65 • create table student2(PRN int);
66 • insert into student2 values(100);
67 • select * from student2;
68 • use Anu1;
69 • show tables;
70 • select * from student2;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	PRN			
	100			

```

65 • create table student2(PRN int);
66 • insert into student2 values(100);
67 • select * from student2;
68 • use Anu1;
69 • show tables;
70 • select * from student2;
71 • truncate table student2;
72 • select * from student2;
73

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
PRN				

```

73 • drop table student2;
74 • use Anu1;
75 • show tables;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
Tables_in_anu1				
▶ s2				
student				

Insert Commands:

1. Insert a Single Row:

```

insert into student (PRN, Name) values(101,"aman");

select * from student;

```

2. Insert Without Specifying Columns

```

insert into student values(101,"aman");

select * from student;

```

3. Insert Multiple Rows at Once

```

11 • insert into student values
12     (102,"Raman"),
13     (103,"Naman"),
14     (104,"Ankit");
15 • select * from student;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	PRN	Name			
▶	101	aman			
	101	aman			
	102	Raman			
	103	Naman			
	104	Ankit			

4. Insert Using SET (Alternative Syntax)

```

16 • insert into student
17     set name = "Vidhi", PRN=105;
18 • select * from student;

```

Result Grid			Filter Rows:
	PRN	Name	
	101	aman	
	101	aman	
	102	Raman	
	103	Naman	
	104	Ankit	
	105	Vidhi	

5.

6. Insert from Another Table

```

19 • insert into student1 (Name, PRN)
20     select Name, PRN
21     FROM student;
22

```

Error Code: 1146. Table 'sales.student1' doesn't exist

```

22 • create table student1 (PRN int, Name varchar(10));
23 • insert into student1 (Name, PRN)
24     select Name, PRN
25     FROM student;
26 • select * from student1
27

```

Result Grid		
	PRN	Name
▶	101	aman
	101	aman
	102	Raman
	103	Naman
	104	Ankit
	105	Vidhi

```

27 • create table student2 (Name varchar(10), PRN int);
28 • insert into student2 (PRN, Name)
29     select Name, PRN
30     FROM student;
31 • select * from student2

```

Output			
#	Time	Action	Message
✓	16 12:11:00	select * from student1 LIMIT 0, 1000	6 row(s) returned
✓	17 12:11:11	create table student2 (Name varchar(10), PRN int)	0 row(s) affected
✗	18 12:11:18	insert into student2 (PRN, Name) select Name, PRN FROM student	Error Code: 1366. Incorrect integer value: 'aman' for column 'PRN' at row 1

```

27 • create table student2 (Name varchar(10), PRN int);
28 • insert into student2 (Name, PRN)
29     select Name, PRN
30     FROM student;
31 • select * from student2;

```

Result Grid		
	Name	PRN
▶	aman	101
	aman	101
	Raman	102
	Naman	103
	Ankit	104
	Vidhi	105

DDL command: Rename Query

RENAME TABLE s2 TO S3;

Show tables;

DML Command: Update

The UPDATE statement is used to modify existing rows in a table. There are **several formats (syntaxes)** depending on what you want to update—single column, multiple columns, using conditions, joins, subqueries, limits, ordering, etc.

1. Basic UPDATE (Single Column)

```
77 • UPDATE student
78   SET s_fee = 210
79   WHERE PRN = 102;
80 • select * from student;
```

Result Grid			
Filter Rows:			
	PRN	name	s_fee
▶	102	Raman	210.000
	103	Naman	300.000
	104	Rohit	100.000
•	NULL	NULL	NULL

2. UPDATE Multiple Columns

```
81 • UPDATE student
82   SET Name=" Ankit",
83       s_fee = 300
84   WHERE PRN = 104;
85 • select * from student;
```

Result Grid			
Filter Rows:			
	PRN	name	s_fee
▶	102	Raman	210.000
	103	Ankit	300.000
	104	Ankit	300.000
•	NULL	NULL	NULL

3. UPDATE Using Arithmetic Expressions

```
UPDATE student
SET s_fee = s_fee* 1.10
WHERE name='Raman';
select * from student;
```

4. UPDATE Without WHERE

```
UPDATE student
SET s_fee = 500;
select * from student;
```

insert into student values

(105,"Neha", 200),

(106,"Riya", 300),

(107,"Yug", 600);

select * from student;

5. UPDATE Using ORDER BY

UPDATE student

SET Name = 'ABC'

ORDER BY s_fee DESC;

select * from student;

6. UPDATE with WHERE IN

UPDATE student

SET name= 'Ankit'

WHERE PRN IN (101, 103);

select * from student;

	PRN	name	s_fee
▶	102	ABC	500.000
	103	Ankit	500.000
	104	ABC	500.000
	105	ABC	200.000
	106	ABC	300.000

Delete

In MySQL, the DELETE statement is used to remove rows from a table. Like UPDATE, it has multiple formats/syntaxes depending on whether you delete:

- specific rows
- all rows
- rows using conditions
- rows using joins
- rows using subqueries
- limited number of rows
- from multiple tables

1. Basic DELETE with WHERE

```
25 • DELETE FROM student
26 WHERE PRN = 105;
27 • select * from student;
```

Result Grid			
Filter Rows:			
	PRN	name	s_fee
▶	102	ABC	500.000
	103	Ankit	500.000
	104	ABC	500.000
	106	ABC	300.000
	107	ABC	600.000

2. DELETE All Rows (Without WHERE)

```
DELETE FROM student;
```

3. DELETE Using ORDER BY

	PRN	name	s_fee
▶	102	ABC	500.000
	103	Ankit	500.000
	104	ABC	500.000
	106	ABC	300.000
	107	ABC	600.000

```
28 • DELETE FROM student
29 ORDER BY name
30 LIMIT 2;
31 • select * from student;
32
```

Result Grid			
Filter Rows:			
	PRN	name	s_fee
▶	103	Ankit	500.000
	106	ABC	300.000
	107	ABC	600.000
*	NULL	NULL	NULL

4. DELETE with IN

```

30 • DELETE FROM student
31 WHERE PRN IN (106, 107);
32 • select * from student;

```

Result Grid			
	PRN	name	s_fee
▶	103	Ankit	500.000
•	NULL	NULL	NULL

Select Query:

1) Select all columns

```
SELECT * FROM table_name;
```

2) Select specific columns

```
SELECT column1, column2
```

```
FROM table_name;
```

SELECT with Filtering & Sorting

3) With WHERE (conditions)

```
SELECT col1, col2
```

```
FROM table
```

```
WHERE condition;
```

DISTINCT

```
SELECT DISTINCT col
```

```
FROM table;
```

→ Removes duplicate rows.

SELECT with expressions

```
SELECT col1, col2 + 10 AS newval
```

```
FROM table;
```

→ You can compute values, rename with AS